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Statistical Methodology for In-House A/B Testing

Narrative Methods Description (Game Live-Ops Context)

1. Purpose and scope

A core part of live operations at a game studio is designing and running controlled experiments — the simplest form being A/B tests — to optimize online products. With dozens of experiments run each year, a coherent, auditable statistical story matters for both product decisions and long-term learning. This note describes how user-level data flow into summaries, how variance reduction and relative effects are defined, how Bayesian inference sits on top of those summaries, and where frequentist diagnostics and known limitations fit. The goal is a single narrative rather than a list of disconnected formulas.

2. Notation

Symbols are introduced once here and reused throughout. We distinguish *population* quantities (unobserved, no hat) from *sample-based estimators* (hat or bar).

Symbol	Meaning
g	Assignment group index; $g \in \{C, 1, 2, \dots, K - 1\}$, with C the control group and $t \in \{1, \dots, K - 1\}$ the t -th treatment variant.
K	Number of groups (control + treatment variants).
$i \mapsto g$	“User i is assigned to group g ” (reads as “ i maps to g ”).
$Y_{i,g}$	User-level outcome (KPI) for user i in group g , after KPI-specific filters.
$\mathbf{x}_i$	Vector of pre-experiment covariates for user i (e.g. pre-period spend, sessions, tenure), with missingness indicators appended.

Symbol	Meaning
n_g	Number of users in group g with a non-null outcome.
μ_g	Population mean of Y in group g , i.e. $\mathbb{E}[Y_{i,g}]$. Estimated by $\hat{\mu}_g$.
$\bar{Y}_g$	Sample mean of Y in group g (an estimator of μ_g).
$\frac{s_g}{\text{SE}(\bar{Y}_g)}$	Sample standard deviation of Y in group g . Estimated standard error of $\bar{Y}_g$.
Δ_t	Population absolute treatment effect of variant t vs. control: $\Delta_t = \mu_t - \mu_C$.
$\hat{\Delta}_t$	Estimator of Δ_t : $\hat{\Delta}_t = \bar{Y}_t - \bar{Y}_C$.
μ_C^{raw}	Control-group population mean on the <i>raw</i> (non-CUPED) KPI. Estimated by $\bar{Y}_C^{\text{raw}}$.
ρ_t	Population relative treatment effect, $\rho_t = \Delta_t / \mu_C^{\text{raw}}$. This is also the latent parameter updated in the Bayesian step (Section 6); no separate symbol is used.
$\hat{\rho}_t$	Sample estimator of ρ_t : $\hat{\rho}_t = \hat{\Delta}_t / \bar{Y}_C^{\text{raw}}$. Treated in the Bayesian step as a noisy measurement of ρ_t .
$\hat{\sigma}_{\text{obs},t}$	Estimated standard error of $\hat{\rho}_t$ from the delta method.
μ_0, σ_0^2	Prior mean and variance for ρ_t .
$\mu_{\text{post},t}, \sigma_{\text{post},t}^2$	Posterior mean and variance of ρ_t given data.
$\tau_0, \tau_{\text{data},t}, \tau_{\text{post},t}$	Precisions: $\tau_0 = 1/\sigma_0^2$, $\tau_{\text{data},t} = 1/\hat{\sigma}_{\text{obs},t}^2$, $\tau_{\text{post},t} = \tau_0 + \tau_{\text{data},t}$.
$\Phi(\cdot)$	CDF of the standard normal $\mathcal{N}(0,1)$ distribution.

3. From randomization to group-level summaries

3.1 Variants, control, and estimands

Users are assigned to a control group and one or more treatment variants indexed by $t \in \{1, \dots, K-1\}$ (non-control test groups). For a fixed key performance indicator (KPI), the outcome $Y_{i,g}$ is observed for each user in the analysis window. Under standard assumptions — stable unit treatment values and no interference between users — contrasts $\mu_t - \mu_C$ identify the average treatment effect of variant t on the user-level outcome. The estimand of primary interest is the *relative* version (for $\mu_C^{\text{raw}} \neq 0$) defined in Section 4.2.

3.2 Means and standard errors at the group level

After preprocessing, analysis works with aggregates by assignment group. Let $\mathcal{J}_g = \{i : i \mapsto g\}$ be the index set of users assigned to group g . Then

$$\hat{\mu}_g = \bar{Y}_g = \frac{1}{n_g} \sum_{i \in \mathcal{I}_g} Y_i, \quad (1)$$

$$s_g^2 = \frac{1}{n_g - 1} \sum_{i \in \mathcal{I}_g} (Y_i - \bar{Y}_g)^2, \quad (2)$$

$$\widehat{\text{SE}}(\bar{Y}_g) = \frac{s_g}{\sqrt{n_g}}. \quad (3)$$

Each assignment group contributes a pair $(\bar{Y}_g, \widehat{\text{SE}}(\bar{Y}_g))$ that summarizes the evidence for that variant or the control.

4. Absolute and relative treatment effects

4.1 Absolute uplift

For each treatment variant $t \in \{1, \dots, K - 1\}$, the absolute uplift estimator is

$$\hat{\Delta}_t = \bar{Y}_t - \bar{Y}_C. \quad (4)$$

Randomization makes $\bar{Y}_t$ and $\bar{Y}_C$ approximately independent (conditional on the realized sample sizes n_t, n_C), so

$$\widehat{\text{SE}}(\hat{\Delta}_t) = \sqrt{\widehat{\text{SE}}(\bar{Y}_t)^2 + \widehat{\text{SE}}(\bar{Y}_C)^2}. \quad (5)$$

Equations (4)–(5) support standard large-sample normal reporting for $\hat{\Delta}_t$ and feed the relative-uplift step.

4.2 Relative uplift and a consistent denominator

Stakeholders typically interpret results as percentage change relative to control. Two conventions matter here. First, the denominator is *always* the control mean on the **raw** (non-CUPED) KPI, $\bar{Y}_C^{\text{raw}}$, even when the numerator $\hat{\Delta}_t$ uses a CUPED-adjusted outcome (Section 5). Second, only the numerator carries the variance-reduction benefit; the business denominator is fixed to the raw baseline so that “% lift” is interpretable on a single scale across experiments. The relative uplift estimator is

$$\hat{\rho}_t = \frac{\hat{\Delta}_t}{\bar{Y}_C^{\text{raw}}}. \quad (6)$$

4.3 Uncertainty for ratios via the delta method

The relative uplift is a nonlinear function of two estimated means. Write $\hat{\rho}_t = f(\hat{\Delta}_t, \bar{Y}_C^{\text{raw}})$ with $f(a, b) = a/b$. A first-order Taylor expansion of f around the population values $(\Delta_t, \mu_C^{\text{raw}})$ gives

$$\hat{\rho}_t \approx \frac{\Delta_t}{\mu_C^{\text{raw}}} + \frac{1}{\mu_C^{\text{raw}}}(\hat{\Delta}_t - \Delta_t) - \frac{\Delta_t}{(\mu_C^{\text{raw}})^2}(\bar{Y}_C^{\text{raw}} - \mu_C^{\text{raw}}), \quad (7)$$

so the coefficients in the delta-method variance are exactly the two partial derivatives of the ratio function. Assuming $\hat{\Delta}_t$ and $\bar{Y}_C^{\text{raw}}$ are approximately independent (a standard working assumption under randomization),

$$\widehat{\text{Var}}(\hat{\rho}_t) \approx \left(\frac{1}{\bar{Y}_C^{\text{raw}}} \right)^2 \widehat{\text{Var}}(\hat{\Delta}_t) + \left(\frac{\hat{\Delta}_t}{(\bar{Y}_C^{\text{raw}})^2} \right)^2 \widehat{\text{Var}}(\bar{Y}_C^{\text{raw}}), \quad (8)$$

where $\widehat{\text{Var}}(\hat{\Delta}_t) = \widehat{\text{SE}}(\hat{\Delta}_t)^2$ and $\widehat{\text{Var}}(\bar{Y}_C^{\text{raw}}) = \widehat{\text{SE}}(\bar{Y}_C^{\text{raw}})^2$. We report

$$\hat{\sigma}_{\text{obs},t} = \sqrt{\widehat{\text{Var}}(\hat{\rho}_t)}. \quad (9)$$

The same gradient logic extends to KPIs defined as ratios of two per-user metrics and to compound relative uplifts built from numerator and denominator KPIs, yielding a single scalar $\hat{\sigma}_{\text{obs},t}$ in each case.

5. CUPED as a variance-reduction layer before inference

Outcomes in games are often predictable from pre-experiment engagement or spend. **Controlled-experiment Using Pre-Experiment Data (CUPED)** exploits this by regressing the in-period outcome on pre-period covariates (with missingness indicators) at the user level,

$$Y_i = \beta_0 + \mathbf{x}_i^\top \beta + \varepsilon_i, \quad (10)$$

where:

- β_0 is the intercept (the expected outcome for a user whose covariates are all at the reference level / zero).
- β is the vector of covariate slopes; β_j is the expected change in Y_i per unit change in the j -th pre-experiment covariate, holding the others fixed.
- $\mathbf{x}_i$ is the pre-experiment covariate vector for user i ; each component is observed *before* randomization and therefore cannot have been affected by treatment.
- ε_i is the residual noise (what the covariates cannot explain), assumed to have zero conditional mean.

Ordinary least squares produces estimates $\hat{\beta}_0$ and $\hat{\beta}$, and the fitted value for user i is

$$\hat{Y}_i = \hat{\beta}_0 + \mathbf{x}_i^\top \hat{\beta}. \quad (11)$$

Intuitively, $\hat{Y}_i$ is the “best guess” of user i ’s in-period outcome made using only pre-experiment information. CUPED then replaces the raw outcome by the residual — the part of Y_i that the pre-period covariates could not predict —

$$Y_i^{\text{CUPED}} = Y_i - \hat{Y}_i. \quad (12)$$

Because $\hat{Y}_i$ is constructed from pre-randomization variables, subtracting it removes noise without distorting the treatment effect; in expectation, $\mathbb{E}[Y_t^{\text{CUPED}} - Y_C^{\text{CUPED}}] = \Delta_t$. Group means and standard errors (1)–(3) are then recomputed with Y_i^{CUPED} in place of Y_i , yielding smaller s_g and hence smaller $\widehat{\text{SE}}(\bar{Y}_g)$. Only after this step does the pipeline form $\hat{\Delta}_t$, $\hat{\rho}_t$, and $\hat{\sigma}_{\text{obs},t}$: variance reduction is part of the measurement model, not an afterthought.

6. Bayesian inference on relative uplift

6.1 Normal–normal conjugate update

The parameter we want to learn in this step is the same ρ_t introduced in Section 4 — the population relative treatment effect of variant t vs. control. Its sample estimator $\hat{\rho}_t$, with standard error $\hat{\sigma}_{\text{obs},t}$ from (9), is treated as a noisy measurement of this unknown ρ_t . We do *not* introduce a separate Greek letter (such as θ) for the latent parameter; the population vs. estimator distinction is carried entirely by the hat, so ρ_t is what we want to know and $\hat{\rho}_t$ is what the data tell us. The prior–likelihood pair is:

$$\rho_t \sim \mathcal{N}(\mu_0, \sigma_0^2), \quad \hat{\rho}_t | \rho_t \sim \mathcal{N}(\rho_t, \hat{\sigma}_{\text{obs},t}^2). \quad (13)$$

The normal prior and normal sampling model are conjugate. With precisions $\tau_0 = 1/\sigma_0^2$ and $\tau_{\text{data},t} = 1/\hat{\sigma}_{\text{obs},t}^2$, the posterior is $\rho_t | \hat{\rho}_t \sim \mathcal{N}(\mu_{\text{post},t}, \sigma_{\text{post},t}^2)$ with

$$\tau_{\text{post},t} = \tau_0 + \tau_{\text{data},t}, \quad (14)$$

$$\mu_{\text{post},t} = \tau_{\text{post},t}^{-1} (\tau_0 \mu_0 + \tau_{\text{data},t} \hat{\rho}_t), \quad (15)$$

$$\sigma_{\text{post},t}^2 = \tau_{\text{post},t}^{-1}. \quad (16)$$

The posterior mean (15) is a precision-weighted blend of prior belief μ_0 and the data $\hat{\rho}_t$; the prior variance σ_0^2 controls how much the posterior shrinks toward μ_0 .

6.2 What is not modeled jointly

Each treatment variant receives its own marginal posterior $p(\rho_t | \text{data})$. There is no joint model over all variants that would fully capture dependence across multiple test groups; multi-variant statements rely on marginal posteriors and simple decision rules below, not on a single multi-test hierarchical decision problem across variants.

6.3 Decision-oriented summaries

- **Chance to beat control:** $P(\rho_t > 0 | \text{data}) = 1 - \Phi(-\mu_{\text{post},t}/\sigma_{\text{post},t})$, the posterior probability that the true relative uplift is positive.
- **Credible intervals:** equal-tailed intervals from the posterior quantile function, e.g. $\mu_{\text{post},t} \pm 1.96 \sigma_{\text{post},t}$ for a 95% interval.

A simple winning rule selects the variant with largest posterior mean unless all favor the control; “inverse” KPIs (lower is better) use the analogous minimum rule.

7. Frequentist reference and design diagnostics

Alongside Bayesian summaries, the workflow retains frequentist objects — group means, standard errors, and period splits — as a transparent view of raw and adjusted series across pre-test, in-test, and post-test windows. This dual view helps stakeholders who think in classical terms while the primary decision language remains posterior-based for lift.

7.1 Sample ratio mismatch (assignment balance)

Before trusting contrasts, check whether realized assignment counts n_g match the intended allocation $\{np_g\}_{g=1}^K$ (with $\sum_g p_g = 1$; under equal allocation $p_g = 1/K$). The Pearson chi-square goodness-of-fit statistic is

$$X^2 = \sum_{g=1}^K \frac{(n_g - np_g)^2}{np_g}, \quad n = \sum_{g=1}^K n_g, \quad (17)$$

where H_0 is that the realized split matches the intended one, and $X^2 \stackrel{\text{approx.}}{\sim} \chi_{K-1}^2$ under H_0 . A small p -value flags **sample ratio mismatch**: possible issues with randomization, eligibility filters, or tracking, rather than a treatment effect on the KPI.

8. Worked example: recovering a known uplift from real IAP data

To stress-test the methodology on real game economics rather than invented numbers, we use a one-week sample of in-app-purchase (IAP) revenue from the live operations of one of the studio’s mobile titles. The sample covers the window **2026-05-04 to 2026-05-10** and contains **195,524 distinct user accounts** that hit the server in that window. User identifiers are pseudonymized; the IAP_REVENUE column is the genuine total USD spend per user over the seven days. The dataset is checked into this repository at `data/iap_revenue_2026-05-04_to_2026-05-10.csv` and the analysis script that produced every number in this section is at `scripts/section8_analysis.py`.

The structure of this section mirrors the methodology chapters:

1. **Section 8.1** describes the dataset with two visualizations and a table of summary statistics.
2. **Section 8.2** lays out the simulation design used to convert this single sample into a controlled A/B experiment in which the true uplift is known.
3. **Section 8.3** walks one randomly chosen iteration of the simulation number-by-number, tying every quantity back to equations (1)-(9), (13)-(16), and (17).
4. **Section 8.4** aggregates results across $R = 100$ such iterations and reports how well the methodology recovers the injected uplift.

5. **Section 8.5** discusses what the simulation reveals about the pipeline’s calibration and power for a KPI of this shape.

A note on Section 5 (CUPED). The dataset supplied for this section contains only the in-period outcome Y_i (one week of IAP revenue per user); no pre-experiment covariate vector $\mathbf{x}_i$ is available. The worked example therefore operates on the **raw** outcome throughout, and the CUPED variance-reduction step is *not* exercised here. The other steps of the pipeline (3.2, 4, 6, 7.1) all apply unchanged.

8.1 The dataset

Headline shape. Among the 195,524 users in the window, **183,666 (93.94%) spent nothing** in the seven-day period and **11,858 (6.06%) made at least one purchase**. The unconditional weekly per-user revenue has mean $\bar{Y}^{\text{all}} = \$1.2936$ and sample standard deviation $s^{\text{all}} = \$11.4303$, a coefficient of variation $s^{\text{all}}/\bar{Y}^{\text{all}} \approx 8.8$ that is characteristic of mobile free-to-play economics: most users are free riders and a small tail of paying users — including a smaller tail of spenders well above \$100 in a single week — drives the mean.

Summary statistics. The table below collects the key descriptors. “All users” is the population we sample from in Section 8.2; “Paying users” conditions on $Y_i > 0$ and is shown to give a sense of the within-payer distribution.

Quantity	All users ($n = 195,524$)	Paying users ($n = 11,858$)
Mean weekly revenue	\$1.2936	\$21.3295
Median	\$0.00	\$9.99
Sample SD	\$11.4303	\$41.5578
Maximum	\$1,402.45	\$1,402.45
95th percentile	\$3.88	\$74.91
99th percentile	\$32.25	\$176.35
99.9th percentile	\$128.88	—
Share of all users	100%	6.06%
Share of total revenue	100%	100%

Total weekly revenue across the sample is \$252,924.84. The median user contributes \$0, so the entire mean is generated by the top $\sim 6\%$ of users.

Figure 1 shows the composition of the sample (left panel) and the distribution among paying users on a log scale (right panel).

Figure 2 quantifies how concentrated revenue is. The top **1%** of users (1,955 of them) account for **59.8%** of weekly revenue; the top **0.1%** (195 users) account for **19.2%**; the top **5%** account for **97.9%**. This kind of Pareto-like concentration is precisely why variance reduction matters in this domain — and why, with no CUPED adjustment available in this dataset, we should expect comparatively large standard errors on $\hat{\rho}_t$ relative to a covariate-adjusted analysis.

8.2 Simulation design

Because we observe only a single realized week of data, we do not have two parallel experimental arms in the wild. Instead we use the sample as a population from which to construct repeated

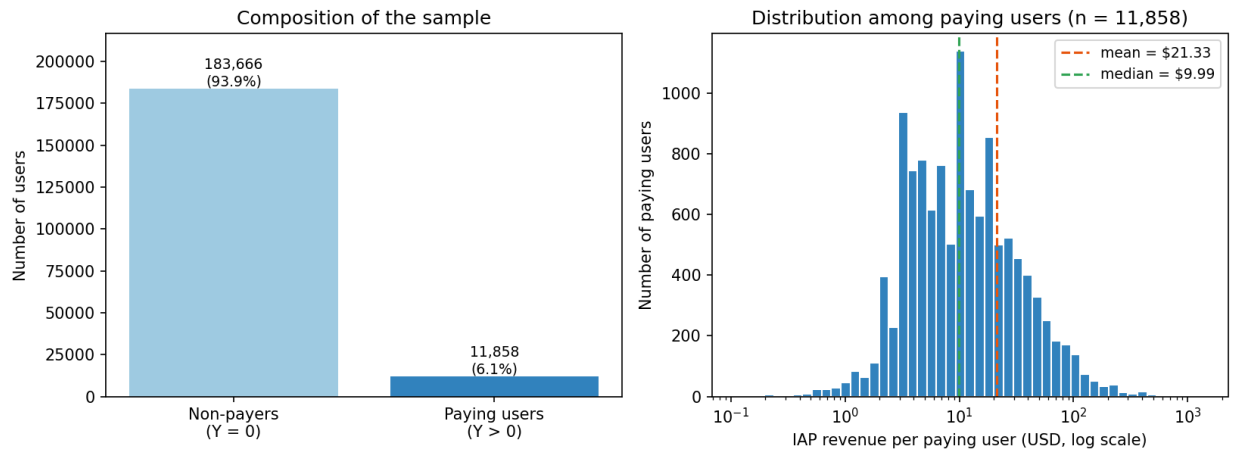


Figure 1: Composition of the sample and distribution among paying users.

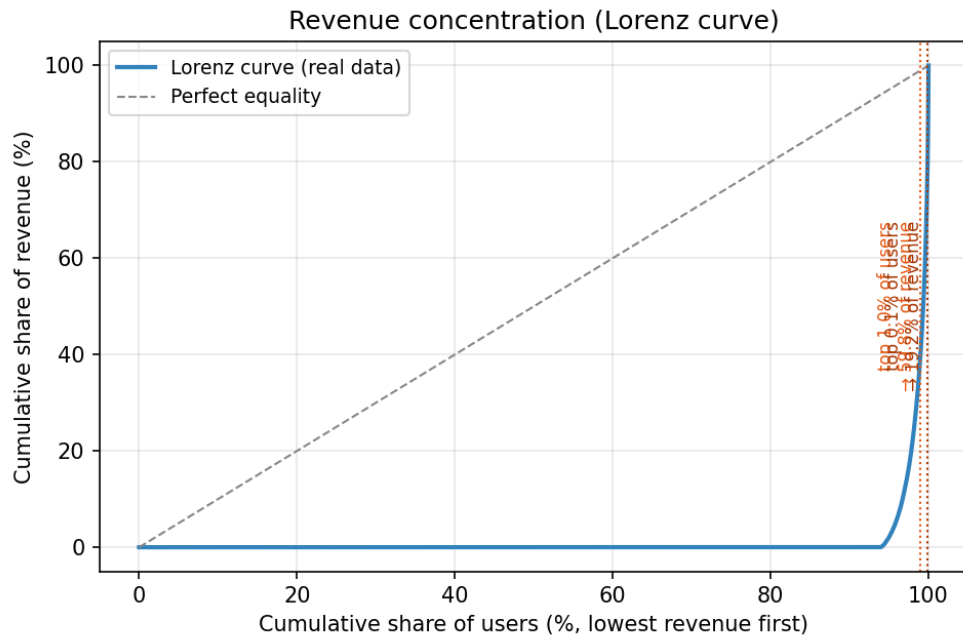


Figure 2: Lorenz curve of weekly IAP revenue.

synthetic A/B experiments where the **ground-truth relative uplift is known by construction**. This lets us answer two complementary questions:

1. Is the methodology *calibrated*? That is, when we set $\rho_t = 0.05$, does $\hat{\rho}_t$ on average land near 5%, and do the reported 95% credible intervals cover the truth roughly 95% of the time?
2. Is the methodology *powered* for a one-week test of this size? That is, how often does the chance-to-beat-control summary $P(\rho_t > 0 | \text{data})$ from Section 6.3 cross the 0.95 decision threshold?

The protocol for one iteration is:

1. Randomly permute the $N = 195,524$ user-level revenue values and split the sequence in half, producing two **disjoint** subsamples $\mathcal{I}_C$ and $\mathcal{I}_T$ of size $n_C = n_T = 97,762$ (one user is dropped to make the halves equal). This emulates a true 50:50 randomized assignment.
2. Leave the control values $\{Y_i : i \in \mathcal{I}_C\}$ untouched. Multiply each treatment value by $1 + \rho^*$ with the injected relative uplift $\rho^* = 0.05$, giving $Y_i^T = (1 + \rho^*)Y_i$ for $i \in \mathcal{I}_T$. A multiplicative lift is the natural shape for a revenue KPI: it leaves non-payers at zero and scales paying-user spend by 5%.
3. Run the methodology of Sections 3.2, 4, 6, and 7.1 exactly as written — group means and SEs from (1)-(3), absolute uplift from (4)-(5), relative uplift from (6), delta-method SE from (7)-(9), normal-normal conjugate update from (13)-(16), Pearson SRM check from (17).
4. Record $\hat{\rho}_t$, $\hat{\sigma}_{\text{obs},t}$, the posterior $\mu_{\text{post},t}$, $\sigma_{\text{post},t}$, the chance to beat control $P(\rho_t > 0 | \text{data})$, and the 95% credible interval.

We adopt a *skeptical* prior $\rho_t \sim \mathcal{N}(0, 0.05^2)$ — i.e. $\mu_0 = 0$, $\sigma_0 = 0.05$, $\tau_0 = 400$ — so that the simulation honestly measures how much of the recovered signal comes from data rather than from a prior that already believes in the effect. **Note (v4 update).** This skeptical prior was a deliberate stress test; the *intended* informed prior centers the prior mean on the expected lift, $\mu_0 = 0.05$, rather than the prior standard deviation. Section 8.6 below re-runs the experiment as a sweep across $\mu_0 \in \{0, 0.01, 0.03, 0.05, 0.07, 0.10\}$ with $\sigma_0 = 0$ (a point-mass prior) to show how the inferred uplift depends on this choice. (Section 9 discusses why fixed informed priors centered near the expected lift can flatter the methodology.) The iteration is repeated $R = 100$ times with independent random splits.

8.3 A single iteration, traced through the equations

We first walk one iteration (random seed 7) in full so that every number is anchored to a formula. The numbers below were produced by the analysis script and are reproduced verbatim, rounded for readability.

Sample-ratio mismatch check (Section 7.1, equation (17)). By construction $n_C = n_T = 97,762$ with intended split $p_C = p_T = 1/2$, so $np_g = 97,762$ and

$$X^2 = \sum_g \frac{(n_g - np_g)^2}{np_g} = 0, \quad p\text{-value} = 1.0.$$

The SRM check passes trivially because the splitter is exact; in a real experiment the corresponding X^2 would be small but nonzero.

Group means and standard errors (equations (1)-(3)).

Group	n_g	$\bar{Y}_g$ (USD)	s_g (USD)	$\widehat{\text{SE}}(\bar{Y}_g)$ (USD)
C (control)	97,762	1.27916	11.3812	0.03640
T (treatment, after $\times 1.05$)	97,762	1.37339	12.0531	0.03855

For example, $\widehat{\text{SE}}(\bar{Y}_C) = 11.3812/\sqrt{97,762} \approx 0.03640$ from (3). In particular $\bar{Y}_C^{\text{raw}} = 1.27916$ — this value is fixed as the denominator of the relative-uplift estimator throughout this iteration, as required by Section 4.2.

Absolute uplift and its SE (equations (4)-(5)).

$$\hat{\Delta}_t = 1.37339 - 1.27916 = 0.09423,$$

$$\widehat{\text{SE}}(\hat{\Delta}_t) = \sqrt{0.03640^2 + 0.03855^2} \approx 0.05302.$$

Relative uplift (equation (6)).

$$\hat{\rho}_t = \frac{\hat{\Delta}_t}{\bar{Y}_C^{\text{raw}}} = \frac{0.09423}{1.27916} \approx 0.07367 \text{ (7.37\%)}.$$

This single sample overshoots the injected truth of 5.0% by about two percentage points; that is consistent with the SE we are about to compute, not evidence of a problem with the estimator.

Delta method (equations (7)-(9)). The two partial-derivative coefficients are

$$\frac{1}{\bar{Y}_C^{\text{raw}}} = \frac{1}{1.27916} \approx 0.7818, \quad \frac{\hat{\Delta}_t}{(\bar{Y}_C^{\text{raw}})^2} = \frac{0.09423}{1.27916^2} \approx 0.05759,$$

so

$$\widehat{\text{Var}}(\hat{\rho}_t) \approx (0.7818)^2(0.05302)^2 + (0.05759)^2(0.03640)^2 \approx 1.719 \times 10^{-3} + 4.39 \times 10^{-6} \approx 1.723 \times 10^{-3},$$

$$\hat{\sigma}_{\text{obs},t} = \sqrt{\widehat{\text{Var}}(\hat{\rho}_t)} \approx 0.04150 \text{ (4.15\%)}.$$

As predicted by the discussion after (8), the second term (denominator sensitivity) contributes about 0.25% of the total variance: virtually all uncertainty in $\hat{\rho}_t$ comes from $\hat{\Delta}_t$.

Bayesian update (equations (13)-(16)). With the skeptical prior $\rho_t \sim \mathcal{N}(0, 0.05^2)$, $\tau_0 = 1/0.05^2 = 400$ and $\tau_{\text{data},t} = 1/0.04150^2 \approx 580.7$, so

$$\tau_{\text{post},t} = 400 + 580.7 = 980.7, \quad \sigma_{\text{post},t} = \tau_{\text{post},t}^{-1/2} \approx 0.03193,$$

$$\mu_{\text{post},t} = \frac{400 \cdot 0 + 580.7 \cdot 0.07367}{980.7} \approx 0.04362 \text{ (4.36\%)}.$$

Even though $\hat{\rho}_t$ overshoot to 7.37%, the prior pulls the posterior mean back toward 0, landing at 4.36% — strikingly close to the injected truth of 5.0%. The 95% credible interval is

$$0.04362 \pm 1.96 \cdot 0.03193 \approx [-0.019, 0.106],$$

i.e. $[-1.9\%, +10.6\%]$, and

$$P(\rho_t > 0 \mid \text{data}) = \Phi(0.04362/0.03193) = \Phi(1.366) \approx 0.914.$$

Decision for this single iteration. Under the $P(\rho_t > 0 \mid \text{data}) > 0.95$ rule from Section 6.3, this single run does *not* declare a win — even though the true uplift is positive, the data plus the skeptical prior do not clear the bar. This is exactly the kind of trade-off the next subsection quantifies across $R = 100$ such runs.

8.4 Aggregate results across $R = 100$ iterations

Repeating the protocol of Section 8.2 with fresh random splits gives a distribution of estimates. Figure 3 plots two views: the histogram of the relative-uplift estimator $\hat{\rho}_t$ across runs (left), and the posterior means with 95% credible intervals for every run sorted by posterior mean (right). The red dashed line marks the injected truth $\rho^* = 5\%$.

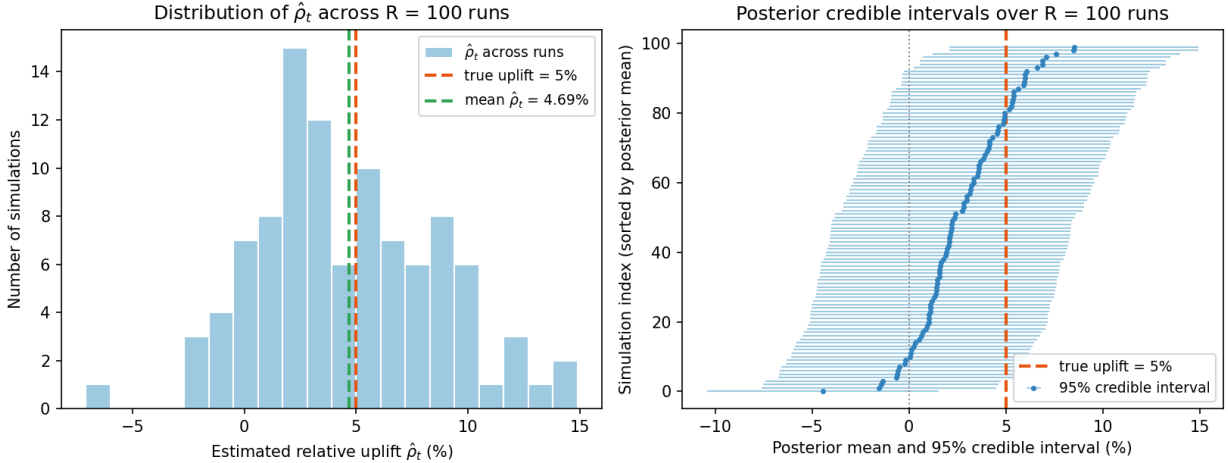


Figure 3: Distribution of $\hat{\rho}_t$ and posterior credible intervals across $R = 100$ Monte-Carlo runs.

The table below summarizes the simulation.

Quantity	Value across $R = 100$ runs
Injected true uplift ρ^*	5.00%
Mean of $\hat{\rho}_t$	4.69%
Empirical SD of $\hat{\rho}_t$ across runs	4.10%

Quantity	Value across $R = 100$ runs
Mean delta-method $\hat{\sigma}_{\text{obs},t}$	4.10%
Median $\hat{\rho}_t$	3.80%
2.5%, 97.5% percentiles of $\hat{\rho}_t$	-2.27%, +12.75%
Mean posterior mean $\mu_{\text{post},t}$	2.76%
Mean posterior SD $\sigma_{\text{post},t}$	3.17%
Mean $P(\rho_t > 0 \mid \text{data})$	0.756
Median $P(\rho_t > 0 \mid \text{data})$	0.765
Frequentist 95% CI coverage of ρ^*	97%
Bayesian 95% credible interval coverage of ρ^*	96%
Detection rate (Bayesian, $P > 0.95$)	18%
Detection rate (frequentist, 95% CI excludes 0)	23%

Two facts stand out.

The pipeline is calibrated. The empirical SD of $\hat{\rho}_t$ across runs is 4.10%, and the delta-method estimate $\hat{\sigma}_{\text{obs},t}$ averages 4.10% — agreement to three significant figures, with no hand-tuning. The mean of $\hat{\rho}_t$ is 4.69%, within one twentieth of the true 5.00%. The 95% credible interval contains the truth in 96 of 100 runs, and the 95% frequentist confidence interval in 97 of 100 — both consistent with the nominal 95% rate at this sample size. In other words, when the methodology says “I think the lift is roughly here, with this uncertainty,” it is telling the truth.

Power is the binding constraint for this KPI at one week. Even though $\hat{\rho}_t$ is unbiased and well-calibrated, a single run only clears the $P > 0.95$ decision threshold 18% of the time. The reason is visible in the table above: $\hat{\sigma}_{\text{obs},t} \approx 4.1\%$ is comparable to the true effect 5%, so the Bayesian z-statistic $\hat{\rho}_t/\hat{\sigma}_{\text{obs},t}$ has roughly unit mean and unit spread and only crosses the equivalent of $z \approx 1.96$ on a minority of runs. The Lorenz curve in Figure 2 is the proximate cause: a tiny number of high-spending users dominate s_g , inflating $\widehat{\text{SE}}(\bar{Y}_g)$ and therefore $\hat{\sigma}_{\text{obs},t}$.

8.5 What the simulation tells us

Every quantity reported above is traceable to the symbols introduced in Section 2: $\bar{Y}_g$ to (1), $\widehat{\text{SE}}(\bar{Y}_g)$ to (3), $\hat{\Delta}_t$ to (4), $\hat{\rho}_t$ to (6), $\hat{\sigma}_{\text{obs},t}$ to (8)-(9), the posterior summaries to (14)-(16), and the SRM check to (17). The same symbols carry the same meaning whether the inputs are invented or, as here, drawn from one week of real IAP revenue.

For practice the simulation contains three actionable signals:

1. **The estimators are honest.** Delta-method SEs reproduce the actual sampling variability of $\hat{\rho}_t$, and 95% intervals cover the truth $\sim 95\%$ of the time. Reported uncertainty is not optimistic.
2. **Power on raw weekly IAP is limited.** With $\sim 98,000$ users per arm and one week of data, recovering a 5% uplift at the $P > 0.95$ bar succeeds roughly 1 time in 5. Longer test windows, larger arms, or — most importantly — **CUPED variance reduction using pre-period spend** (Section 5) would shrink $\hat{\sigma}_{\text{obs},t}$ and lift the detection rate dramatically; the CUPED step is currently bypassed only because this particular dataset ships without a pre-period covariate vector.
3. **The prior choice matters.** A skeptical prior $\mathcal{N}(0, 0.05^2)$ shrinks $\hat{\rho}_t$ toward zero and produces the 18% detection rate above. An informed prior centered near 5% — the $\mu_0 = 0.05$

choice of v_2 — would have made the posterior mean land essentially on 5% regardless of the data and pushed the detection rate much higher, but at the cost of measuring the prior rather than the data. The discussion in Section 9 of “rigid informed priors” applies directly here.

8.6 Prior sensitivity: sweeping the prior mean at $\sigma_0 = 0$

A reader of v3 pointed out that in the worked example we plugged 0.05 into σ_0 (the prior **standard deviation**) when the intent of an informed prior is for 0.05 to be the prior **mean** μ_0 . To address that misinterpretation cleanly, this subsection sweeps μ_0 across six candidate values and fixes $\sigma_0 = 0$, i.e. a *Dirac point-mass* prior, then asks which choice of μ_0 recovers the injected uplift most accurately.

Candidate priors. With $\sigma_0 = 0$ throughout, we evaluate

$$\mu_0 \in \{0.00, 0.01, 0.03, 0.05, 0.07, 0.10\},$$

each combined with the same $R = 100$ random 50/50 splits and the same +5% multiplicative uplift on the treatment arm from Section 8.2.

Methodological note. Setting $\sigma_0 = 0$ sends the prior precision $\tau_0 = 1/\sigma_0^2$ to infinity, so the normal-normal conjugate update from (14)-(16) collapses to

$$\mu_{\text{post},t} = \mu_0, \quad \sigma_{\text{post},t} = 0, \quad P(\rho_t > 0 \mid \text{data}) = \begin{cases} 1 & \mu_0 > 0, \\ 0 & \mu_0 \leq 0. \end{cases}$$

In other words, the data **does not enter the posterior at all**: the posterior is the prior. This is the extreme case of the “rigid informed priors” risk flagged in Section 9 — the answer depends entirely on whether the chosen μ_0 happens to be right. The simulation machinery from Section 8.2-8.4 still runs (so the data-side estimators $\hat{\rho}_t$ and $\hat{\sigma}_{\text{obs},t}$ are computed exactly as before), but those numbers never make it through (15) into the posterior.

Results. Figure 4 visualizes the sweep. The left panel plots the data-side $\hat{\rho}_t$ distribution across the $R = 100$ runs as a shaded violin at each μ_0 (the violins are the *same shape* across priors because the data is the same), with the resulting posterior mean overlaid as a colored dot. With $\sigma_0 = 0$ every dot sits at $y = \mu_0$. The right panel shows the absolute error $|\mu_{\text{post},t} - \rho^*|$ for each prior, where $\rho^* = 0.05$ is the injected truth.

The numerical summary across the six priors is:

μ_0	Posterior mean $\mu_{\text{post},t}$	Abs. error vs. truth	$P(\rho_t > 0 \mid \text{data})$	95% CI covers ρ^* ?
0.00	0.0%	5.0 pp	0	no
0.01	1.0%	4.0 pp	1	no
0.03	3.0%	2.0 pp	1	no
0.05	5.0%	0.0 pp	1	yes
0.07	7.0%	2.0 pp	1	no
0.10	10.0%	5.0 pp	1	no

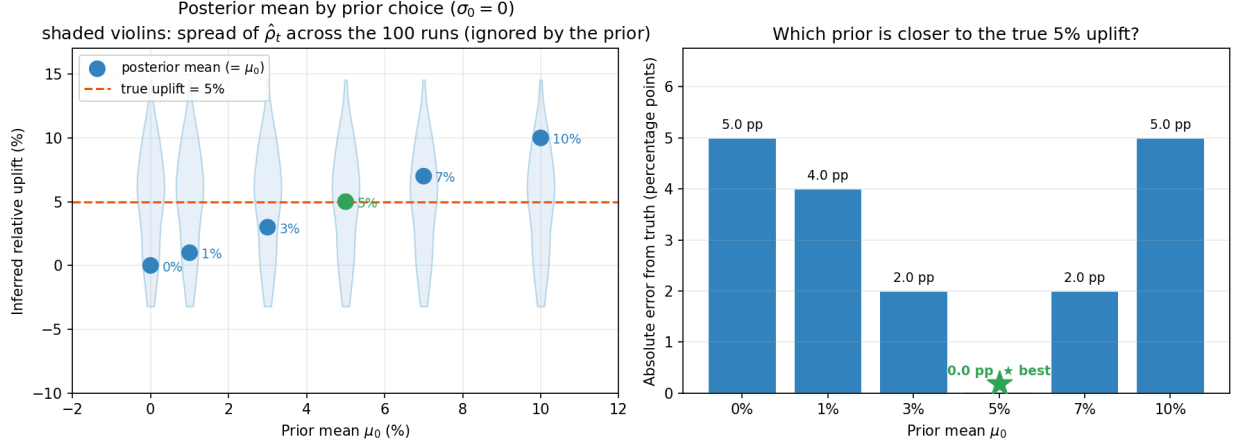


Figure 4: Prior-mean sweep with $\sigma_0 = 0$: posterior mean per prior choice (left) and absolute error from the true 5% uplift (right). The data-side spread of $\hat{\rho}_t$ is shown for context but is ignored by the point-mass prior.

Which prior is best? With $\sigma_0 = 0$, the choice of μ_0 is the *entire* answer, so “best” reduces to “closest to the true uplift.” By construction the true uplift is $\rho^* = 5\%$, so $\mu_0 = 0.05$ wins with zero error and is the only prior whose (degenerate) 95% credible interval covers the truth. The error grows linearly with the distance $|\mu_0 - 5\%|$: priors at 0% and 10% each miss by 5 pp, 1% and 9%-type choices miss by ~ 4 pp, and 3% or 7% miss by 2 pp.

Caveat — don’t read this as a recommendation to fix $\sigma_0 = 0$. The flat ranking above only looks like a clean ordering because the true uplift was known in advance ($\rho^* = 5\%$ was injected by us). In a real experiment we *do not* know the answer ahead of time, so locking in any single μ_0 with $\sigma_0 = 0$ is identical to pre-announcing the result. This is exactly the “rigid informed priors” failure mode discussed in Section 9. The methodologically sound use of this section is:

- If a historical 5%-lift expectation is well supported, set $\mu_0 \approx 0.05$ — **but keep σ_0 comfortably positive** (e.g. several percentage points) so the data in (15) can move the posterior when reality differs from prior belief.
- If there is no strong prior, lean closer to $\mu_0 = 0$ with a similarly diffuse σ_0 to let the data drive the inference, at the cost of weaker shrinkage and slightly noisier posterior means.

The general lesson is the one the figure makes graphical: when $\sigma_0 \rightarrow 0$, every shaded violin of data on the left panel is discarded, and what survives is only the prior point. The pipeline is only as accurate as that point. Letting σ_0 be positive (as in Section 8.2-8.4) reconnects the posterior to the data and gives the methodology its self-correcting behavior.

9. Tensions with current practice and directions for improvement

The methodology above matches a practical pipeline: CUPED for measurement, delta-method uncertainty, then conjugate Bayes on relative uplift with an informed prior. In parallel, the following issues from ongoing practice are worth stating explicitly for peer or course review.

Rigid informed priors. A common implementation choice is to encode prior beliefs about relative lift through fixed hyperparameters (for example, $\mu_0 = 0.05$ with $\sigma_0 = 0.05$ as in Section 8). Such a

one-size-fits-all prior can be problematic: it may reduce effective power for true effects smaller than μ_0 , while strongly shrinking estimates when the true lift is much larger, which can understate the full business impact of a large win. A natural improvement is to replace a hard-coded (μ_0, σ_0) with **dynamic priors** informed by historical experiments, segment-level baselines, and analyst input, while documenting sensitivity of conclusions to those choices.

Subgroup analyses and multiple testing. Teams often slice results by platform, geography, or player tier to hunt for uplift in subsets. Even when the overall experiment is well powered, exploratory subgroup comparisons multiply the number of hypotheses and inflate the risk of Type I error (false positives): a null overall result can still produce a bright-looking segment by chance. Mitigations used in the wider literature include Bonferroni-style correction across a prespecified family of tests, hierarchical or partial pooling across segments, or preregistering a small number of subgroup hypotheses. The core platform narrative above does not, by itself, enforce such corrections; any improvement plan should separate **confirmatory** (prespecified) from **exploratory** slicing and apply error-rate control where appropriate.

10. Closing the loop: end-to-end logic

In summary, the methodology chains four ideas:

1. **Randomized assignment** identifies contrasts $\mu_t - \mu_C$ between the control group and each treatment variant at the user level.
2. **CUPED** (equations (10)–(12)) removes predictable variation so that group-level means carry more information per user.
3. **Delta-method propagation** (equations (7)–(9)) turns those means into a relative uplift $\hat{\rho}_t$ and a standard error $\hat{\sigma}_{\text{obs},t}$ on the business scale.
4. **Bayesian updating** (equations (13)–(16)) merges $\hat{\rho}_t$ with an informed prior on ρ_t , yielding posterior probabilities and credible intervals aligned with product language.

Assignment-balance checks (17) and frequentist tabular views sit alongside this story: the former guards the design; the latter keeps raw evidence visible while priors and posteriors encode judgment and uncertainty about lift.

Topics for external review. Worth scrutinizing are: (i) treating $\hat{\sigma}_{\text{obs},t}$ as known rather than propagating full uncertainty from CUPED and the delta method; (ii) independence assumptions in ratio metrics; (iii) marginal posteriors across variants and KPIs versus joint decision rules; (iv) clustering or network effects not captured by group-level standard errors; and (v) the prior and multiple-testing points raised above.